

SGE: Semantically-Guided Exploration for Unstructured Environments via Image-Space Waypoint Sampling

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Abstract—This work introduces Semantically-Guided Exploration (SGE), a modular exploration framework for ground vehicles that integrates pixel-level semantic segmentation into sampling-based waypoint selection and receding-horizon route optimization. Unlike conventional geometric exploration methods, SGE evaluates candidate exploration goals directly in the image space using a semantic-aware utility function that accounts for terrain traversability, obstacle proximity, objects of interest, and depth-based exploration reward. Sampled waypoints are projected into 3D and ordered through a real-time Traveling Salesman Problem (TSP) formulation, enabling receding-horizon goal selection. To address real-world navigation uncertainty, the framework introduces mechanisms, including temporary taboo regions to handle navigation failures and a graph-based relocation strategy for efficient backtracking across explored areas. We evaluate SGE in standardized simulation benchmarks against state-of-the-art exploration planners and demonstrate competitive performance in volumetric coverage, while enabling semantic task biasing that cannot be achieved by purely geometric methods. The framework is further validated through real-world experiments using multiple robotic platforms in indoor campus buildings and in limestone and coal mines. Results show consistent performance and adaptability across platforms and domains.

I. INTRODUCTION

Exploration robots play a critical role across many domains. In space and deep-sea missions, they enable scientific discovery in environments inaccessible to humans [1], [2]. In underground environments and search-and-rescue scenarios following natural disasters, robots can operate where conditions are unsafe or impractical for human responders [3]. Robots are also increasingly used for environmental monitoring, supporting conservation efforts related to climate change and wildlife protection [4]. Across these applications, environments are often hazardous, and communication is unreliable or non-existent, making teleoperation infeasible and autonomous robotic exploration essential.

Autonomous robot exploration has advanced significantly in recent years, with systems capable of long-duration operation and accurate mapping in challenging environments, including underground settings [5]. Nevertheless, most existing exploration algorithms rely primarily on geometric information, such as occupancy, distance, or frontier-based metrics, which are a restricted subset of the rich information available from onboard sensors. Humans, in contrast, leverage prior experience and semantic understanding of a scene when deciding where to go next [6]. Cues such as terrain type, signage, and task-relevant objects are already observable by the

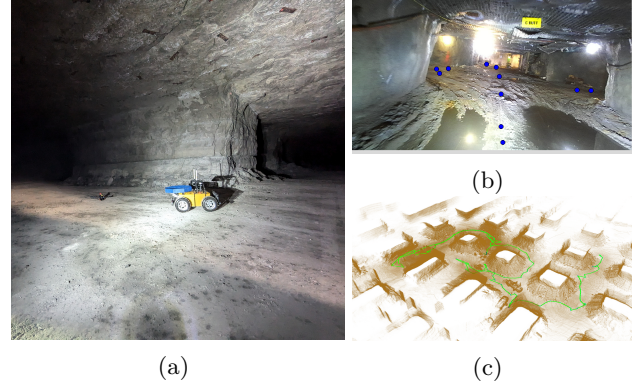


Fig. 1: SGE deployed in underground mines. (a) The Rhino robot exploring a limestone mine. (b) Onboard view during coal-mine exploration: image-space waypoints (blue) sampled from the semantic utility distribution avoid regions classified as mud and water. (c) Resulting 3D reconstruction and executed trajectory (green) in the limestone mine.

robot’s camera. This work exploits them directly at the point where exploration goals are generated.

This research aims to enhance autonomous robotic exploration by integrating semantic perception directly into the waypoint selection process of sampling-based planners. We introduce Semantically-Guided Exploration (SGE), a modular framework for unmanned ground vehicles that incorporates pixel-level semantic reasoning into real-time goal generation while preserving the computational efficiency of classical exploration methods. Rather than replacing geometric planning, SGE augments it with semantic-aware importance sampling and receding-horizon route optimization, enabling flexible and task-adaptive behavior across diverse environments. Figure 1 shows the robot platform described in [7] performing SGE in an underground limestone mine, together with an example of onboard semantic waypoint sampling and exploration environment reconstruction.

The contributions of this work are:

- 1) Semantic-weighted waypoint sampling in image space, where candidate waypoints are generated through a utility-driven formulation that incorporates terrain traversability, obstacle proximity, task-relevant objects, and depth-based exploration incentives. This lightweight formulation enables real-time operation without constructing dense se-

semantic 3D maps.

- 2) A receding-horizon TSP-based exploration manager that continuously optimizes candidate waypoints while maintaining scalability for online deployment.
- 3) Robust uncertainty-handling mechanisms, including temporary taboo regions for managing navigation failures and a graph-based relocation strategy that enables efficient backtracking across previously explored areas without global replanning.
- 4) Extensive validation across simulation and in real-world deployments with two robotic platforms (Rhino and Husky ground vehicles) in diverse environments such as indoor campus buildings, limestone and coal mines.

The remainder of this paper is organized as follows. Section II reviews related work in robotic exploration. Section III presents the proposed SGE framework, including semantic-based waypoint selection and the exploration manager. Section IV evaluates the approach in simulation and real-world experiments and compares it with state-of-the-art methods. Finally, Section V concludes the paper and outlines directions for future research.

II. Related Work

Autonomous robot exploration has advanced significantly in recent years. The DARPA Subterranean Challenge (SubT) [8] drove many recent advances by requiring ground and aerial robots to autonomously explore challenging underground settings. Similarly, the Autonomous Exploration Development Environment [9] provides a standardized benchmark for evaluating exploration algorithms under controlled conditions.

Existing methods can broadly be categorized as information-based, frontier-based, or sampling-based. Information-based methods [10], [11] select actions that maximize expected information gain at each robot action; [12] and [13] introduce a Cauchy-Schwarz Quadratic Mutual Information formulation that increases efficiency, enabling higher resolution 3D voxel grids. Frontier-based methods [14], [15], [16] drive the robot toward the frontier, which is the boundary between known and unknown space; [17] samples candidate frontiers and orders them by TSP. Sampling-based planners evaluate subsets of the environment at each time step: NBVP [18] applies a receding horizon over *RRT** generated viewpoints, GBPlanner [19] uses rapidly exploring random graphs with a global layer for repositioning, and [20], [21], [22] extend the paradigm with motion primitives, model predictive control, and continuous tree rewiring, respectively. Hybrid methods combine paradigms: DSVF [23] alternates frontier- and sample-based stages, and TARE [24] couples a local viewpoint planner with global TSP optimization and is currently the best-performing method in the Autonomous Exploration Development Environment benchmark. All of these approaches reason

over geometric representations derived from LiDAR or occupancy maps. When different terrain types share similar geometry (e.g., mud versus solid ground) or when visual cues such as signage or task-relevant landmarks are informative, purely geometric reasoning may not fully exploit available perceptual information.

Initial efforts to incorporate semantics into exploration differ in where semantics enters the pipeline. A first family builds a semantic 3D map and plans over it. Dang et al. [25] annotate an incrementally built map with visual saliency and bias a receding-horizon sampling planner toward salient regions; [26] maintains a semantic octree and derives semantic information gain from them; and [27] combines volumetric exploration with inspection or object-centric behaviors around semantic entities. These methods obtain rich global reasoning, but semantics influences planning only after it has been fused into the map, incurring the cost and latency of dense semantic reconstruction. A second family keeps geometric candidate generation and uses semantics to re-score or search among candidates: [5] integrates object detections into an object-search behavior; [28] clusters unexplored point-cloud regions into a topological graph that guides high-level decisions; [29] scores next-best-view poses by the semantic features visible from them; and STAGE [30] shifts the endpoint of each local path segment toward the most traversable pixel of a semantic traversability image, applying semantics as a local corrective on paths toward geometrically selected goals. Because candidates are still generated geometrically, these methods cannot propose goals that only semantics would distinguish, and effort is spent evaluating candidates that terrain semantics would have excluded outright. SGE differs from both in that pixel-level semantics is the primary driver of candidate generation: waypoints are sampled directly from a semantic utility distribution over the image plane and only then projected to 3D and validated geometrically. Semantic influence is applied at bounded per-frame cost with no dense semantic map, which matters for real-time operation on embedded platforms; goal generation requires only an RGB-D camera, and task-conditioning is achieved by adjusting class weights in the utility function rather than by modifying the planner.

III. Methodology

The proposed SGE framework consists of two main components, as illustrated in the architecture diagram in Figure 2: the Waypoint Selection module, which samples candidate waypoints from semantic image inputs, and the Exploration Manager, which aggregates contextual information and selects the robot's next goal in a receding-horizon manner. The decision-making process is continuously updated until full environment coverage is achieved.

The sampling process is guided by semantically segmented images from the robot's camera, identifying

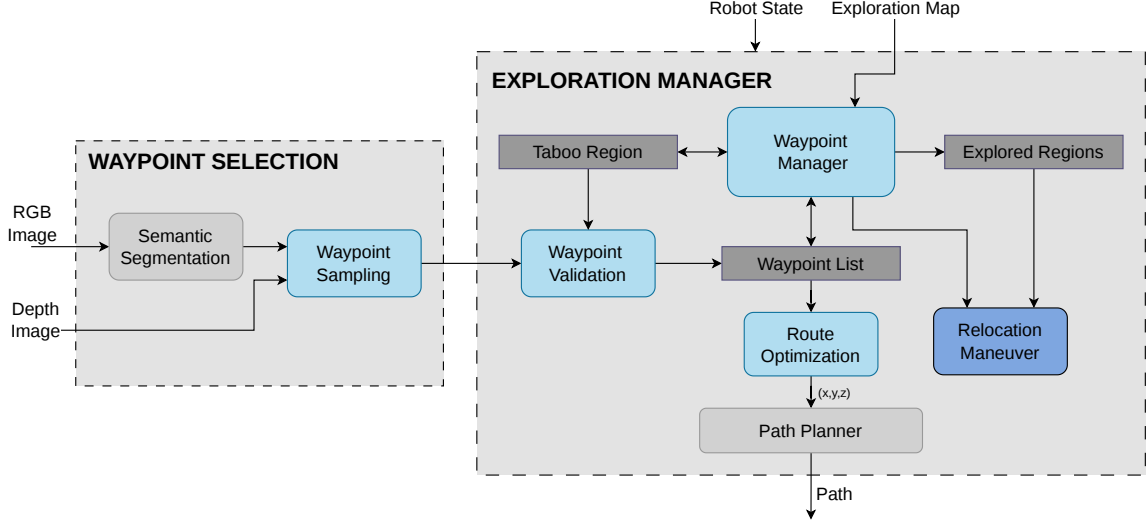


Fig. 2: Semantic-Guided Exploration (SGE) architecture diagram. Light blue blocks are synchronous modules, dark blue blocks represent asynchronous modules, and light gray blocks correspond to third-party algorithms. Dark gray blocks indicate runtime-evolving data structures whose size changes during exploration.

different terrain types and obstacles. Candidate waypoints are ordered by solving a TSP starting from the robot’s current position, and the next waypoint is selected as the first element of the optimized sequence. As new waypoints are generated, the list is continuously updated by adding new candidates and removing visited waypoints. In real-world deployment, exploration is subject to uncertainty: navigation failures may occur, some waypoints may be unreachable due to obstacles, the explored region continuously grows, and the number of candidate exploration goals must remain bounded to ensure real-time TSP optimization [31]. To address these challenges, the framework introduces taboo regions and relocation maneuvers, described in detail in the following sections.

A. Semantically-guided Waypoint Selection

The waypoint selection module generates candidate exploration goals directly from semantically segmented RGB images and aligned depth measurements, as shown in Figure 3. At each time step, the segmentation network produces a pixel-level classification map, from which traversable regions, obstacles, and optional task-relevant objects are identified. Rather than constructing a dense semantic 3D map, waypoint selection is performed directly in the image plane, enabling lightweight real-time operation.

We formulate the waypoint selection process as a semantic-weighted sampling problem defined over the image plane. Rather than uniformly sampling traversable pixels or relying solely on geometric frontier detection, we assign each pixel i a utility score $U(i)$ that reflects its estimated exploration value based on semantic and geometric cues. This formulation enables lightweight

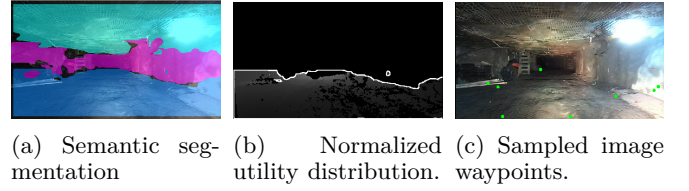


Fig. 3: Semantic-guided waypoint sampling process. (a) Semantic segmentation of the RGB input. (b) Utility-based sampling distribution derived from (1). (c) Candidate waypoints, shown as green dots, selected in the image plane prior to 3D projection.

evaluation directly in image space, avoiding the need for dense semantic volumetric reconstruction while preserving real-time performance.

For pixels classified as traversable, the utility function is defined as:

$$U(i) = \max \left\{ c_a \cdot k_i - \sum_{j=1}^N \left(\frac{c_{bj}}{D_{ij}^2} \right) + e^{1-c_c \cdot D_l(i)} + c_d \cdot z_i, 0 \right\} \quad (1)$$

where each term encodes a distinct exploration incentive or penalty. The rectification operator ensures non-negative utilities prior to normalization into a sampling distribution.

The resulting utility values are normalized to form a probability distribution over candidate pixels:

$$P(i) = \frac{U(i)}{\sum_{m \in I} U(m)} \quad (2)$$

where I denotes the set of all pixels in the image. Waypoints are then sampled from $P(i)$, and only the

selected pixels are projected into 3D space.

The utility function has four complementary terms:

- **Neighboring Traversable Pixels ($c_a k_i$):** The term $k_i \in [0, 8]$ is the count of neighboring pixels classified as traversable. It promotes sampling within traversable regions by favoring pixels surrounded by traversable regions. As a result, it reduces sensitivity to segmentation noise and discourages sampling near semantic boundaries that may correspond to partial obstacles or uncertain classifications. The coefficient c_a controls the relative importance of this spatial consistency term and can be adjusted per terrain type if desired.
- **Distance from Obstacles ($-\sum_{j=1}^N \frac{c_{b_j}}{D_{ij}^2}$):** This inverse-squared penalty discourages sampling near obstacle classes. Here, D_{ij} represents the 2D quasi-Euclidean distance from pixel i to obstacle class j , and c_{b_j} is the penalty weight for each obstacle type. The inverse-squared formulation ensures strong repulsion near obstacles with smooth spatial decay. Distances are computed using a 2D distance transform inspired by TSDF methods [32] and implemented via a grassfire algorithm [33], resulting in $O(V)$ computational complexity, where V is the size of the grid. If an obstacle class is not present in the image, its corresponding distance is set to ∞ , contributing no penalty.
- **Proximity to Objects of Interest ($e^{1-c_c D_I(i)}$):** This term rewards sampling near task-relevant objects (e.g., directional signs like exit markers), where $D_I(i)$ denotes the 2D quasi-Euclidean distance from pixel i to the nearest object-of-interest class. The exponential form provides stronger emphasis at close range while preserving smooth decay with increasing distance. The coefficient c_c regulates the strength of this semantic bias. If no object of interest is detected, $D_I(i)$ is set to ∞ , and the term does not influence the utility.
- **Depth-Based Exploration Reward ($c_d z_i$):** The term z_i denotes the depth at traversable pixel i . This component favors distant traversable regions, encouraging outward expansion of the explored area. It acts as a surrogate for information gain by promoting waypoints that are likely to reveal new space without explicit volumetric entropy computation. The coefficient c_d determines the influence of depth-based expansion relative to other terms.

The weighting coefficients c_a, c_b, c_c, c_d allow task-specific tuning and were selected empirically to balance safety and exploration aggressiveness. An ablation study in Section IV quantifies the contribution of each utility term and manager mechanism. To account for segmentation uncertainty, non-traversable but non-obstacle classes receive a small baseline utility, and obstacle pixels zero probability.

After constructing the probability distribution over

image pixels, a fixed number of indices (u, v) are sampled. The complexity of this operation is $O(n + m \log(n))$, where n is the number of pixels in the discrete distribution and m is the number of sampled points, which reduces to approximately $O(n)$ in practice, since m is small. Each sampled pixel is projected into the camera coordinate frame using the aligned depth image: $z = z_{\text{depth_image}}$, $x = (u - c_x) \cdot z / f_x$, and $y = (v - c_y) \cdot z / f_y$, where f_x and f_y are the focal lengths and c_x and c_y the principal point coordinates. The resulting 3D points are then transformed into the global frame and passed to the Exploration Manager for validation and optimization.

B. Exploration Manager

The Exploration Manager coordinates the overall exploration process and consists of two concurrently running loops. The optimization loop selects goals in a receding-horizon manner, validates incoming waypoints, and maintains the optimized goal list. The navigation loop monitors execution, detects navigation failures, updates taboo regions, and triggers relocation behavior when necessary.

The optimization loop is outlined in Algorithm 1. This algorithm must be fast and scalable to avoid becoming a bottleneck for the navigation loop, where the robot would otherwise have to stop during exploration and wait for new waypoints. The method begins by updating the robot state and sampling new waypoints from the waypoint selection node. Then the Validate Waypoint method validates each new point through a sequence of checks:

- **Boundary check** to ensure the point lies within a predefined exploration region.
- **Traversability check** to verify that the point is within traversable terrain using a local traversability map, a fixed-size region around the robot's current position. Collision checking here is $O(n)$ in complexity, where n represents the number of points in the traversable region map. This check leverages LiDAR data to enhance semantic segmentation, improving the safety of autonomous navigation.
- **Taboo list check** to verify that a waypoint is not inside a list of temporary forbidden locations. This list, although variable in size, is significantly smaller than other data structures. This is implemented by checking whether a sphere of radius r centered at the waypoint intersects any entry in the taboo list.
- **Exploration region check** to confirm that the new point has not yet been explored. Explored regions are represented by a voxel-based structure, where determining if a 3D point falls within a voxel is a constant-time operation, $O(1)$. While this approach allows efficient checks, it does require fixed-size memory allocation.

Valid waypoints are added to a goal list. The next waypoint is selected by continuously solving a TSP with no return, with the robot's current global position as the

Algorithm 1 Exploration Manager Optimization Loop

```

1: while WaypointList  $\neq \emptyset$  do
2:    $p \leftarrow \text{UpdateRobotState}()$ 
3:    $w \leftarrow \text{SampleWaypoint}(p) \triangleright$  Semantic-based sampling
4:    $isValid \leftarrow \text{ValidateWaypoint}(w, \text{TraversabilityMap},$ 
        $\text{TabooList}, \text{ExploredRegion})$ 
5:   if  $isValid$  then
6:      $\text{AddWaypoint}(w)$ 
7:   end if
8:    $D \leftarrow \text{UpdateDistanceMatrix}(\text{WaypointList}, p)$ 
9:    $\text{OrderedWaypointList} \leftarrow \text{SolveTSP}(\text{WaypointList}, D)$ 
10: end while

```

starting point. Euclidean distances between waypoints are used to calculate the TSP cost, with the cost matrix updated at each step before optimization. The complexity of evaluating the full cost matrix is $O(n^2)$, where n is the number of nodes; however, at each step, only the costs associated with newly added or removed nodes are recalculated. The robot’s new distance to all nodes, relative to its current global position, is also computed.

The default SGE behavior is not to consider the cost of returning to the starting position in the optimization by setting the return-to-depot cost to zero. Alternatively, a return cost to $(0,0,0)$ location can be set, where the robot would consider the cost of returning to the start at each optimization step. The TSP is solved using the Guided Local Search (GLS) metaheuristic [34], implemented in OR-Tools [35], which provides efficient results with minimal variability in performance and supports real-time receding horizon operation with up to 100 nodes, as shown in [31].

The Navigation loop is shown in Algorithm 2. This loop monitors robot navigation, builds the exploration map, updates taboo regions, triggers relocation behavior, and removes waypoints once explored. Exploration is completed when no valid waypoints remain. This assumes that the robot has a local planner and path follower algorithm capable of reaching goals and maintaining localization. The exploration map is constructed using registered LiDAR point clouds around the robot. At each step, points are marked as occupied in a voxel-based map, which is an $O(n)$ operation, where n is the number of points in the registered scan.

Taboo regions model temporary navigation failures and uncertainties in the robot’s actions. Inspired by Tabu Search metaheuristic [36], they provide memory of temporarily forbidden areas. When a waypoint is sent to the local planner, the Exploration Manager estimates the traversal time from planned path length and robot velocity, and sets a threshold at 150% of that estimate. The manager monitors the navigation at each loop. If the threshold is exceeded, the region around that waypoint is added to the taboo region. This temporary ban lasts $c \times (t_{planned} - t'_{planned})^2$, where c is a constant value, $t_{planned}$ is the estimated time when the waypoint was set, and

Algorithm 2 Exploration Manager – Navigation Loop

```

1: while  $\text{OrderedWaypointList} \neq \emptyset$  do
2:    $\text{SendNavigationGoal}(w)$ 
3:    $p \leftarrow \text{UpdateRobotState}()$ 
4:    $\text{Exp} \leftarrow \text{UpdateExploredRegion}(p, \text{Map})$ 
5:    $V, E \leftarrow \text{UpdateRelocationManeuverGraph}(p)$ 
6:   if  $\|w - p\| < \tau$  or  $expired$  then
7:      $w \leftarrow \text{PopNextWaypoint}(W)$ 
8:      $\text{ResetTimer}()$ 
9:      $T_{est} \leftarrow \text{EstimateTraverseTime}()$ 
10:     $expired \leftarrow \text{False}$ 
11:   end if
12:   for  $w_t \in \text{TabuList}$  do
13:     if  $w_t.timer < 0$  then
14:        $\text{RemoveFromTabuList}(t)$ 
15:     end if
16:   end for
17:   if  $T_{nav} > T_{est}$  then
18:      $expired \leftarrow \text{True}$ 
19:      $\text{UpdateTabuList}(w, T_{estimated})$ 
20:   end if
21:   if  $\|w - p\| > \tau_r$  then
22:      $\text{RelocationManeuver}(w, V, E)$ 
23:   end if
24: end while

```

$t'_{planned}$ is the remaining estimated time. Entries are removed when their timers reach zero, so these regions are only temporarily avoided.

The relocation maneuver is a strategy to safely and efficiently navigate previously explored regions by backtracking along a graph of traversed positions. This process is based on a bidirectional graph $G = (V, E)$ that represents the regions the robot has already visited, where V is the set of nodes corresponding to these positions and E represents edges between these nodes. The graph G is dynamically updated as the robot moves: if the robot’s current position c exceeds a threshold distance d_{thresh} from any existing node in V , a new node v_n is added to V , along with edges connecting v_n to nearby nodes in V within range $d_{connect}$.

The relocation maneuver is triggered when the goal position g is far from the robot’s current location c as determined by the TSP optimization. The first step in this process is to identify the nearest node $v_s \in V$ to the robot’s current position c and the nearest node $v_g \in V$ to the target position g . If $v_s = v_g$, the goal is already the nearest position from the visited region and the relocation maneuver is not necessary. If $v_s \neq v_g$, the A* algorithm [37] is used to compute the shortest path P in G from v_s to v_g . Euclidean distance is used as the A* heuristic to estimate the cost. Once the path $P = \{v_s, v_1, v_2, \dots, v_g\}$ is computed, the robot sequentially navigates to each node in P until it reaches v_g . During the relocation, the Exploration Manager will

pause other exploration tasks, waiting for the robot to safely complete the backtracking steps through known terrain.

IV. Experiments and Results

To evaluate the SGE framework, we conducted simulation benchmarks against state-of-the-art exploration planners and validated the approach in real-world deployments. The experiments were designed to assess (i) volumetric exploration performance, (ii) behavioral characteristics of image-based sampling, and (iii) semantic task-conditioning capability.

Unless otherwise specified, parameters were fixed across experiments to demonstrate robustness across environments. The utility weights were set to $c_a = 3$, $c_b = 5$, $c_c = 0.2$, and $c_d = 8$, with eight waypoints sampled per time step. The taboo region was defined as a 64m^3 region, and relocation was triggered for goals located more than 20m from the current robot position. These parameters were selected empirically during development to balance safety, exploration aggressiveness, and semantic biasing. The same parameter set was used unchanged across all simulation and real-world experiments, spanning different platforms, sensors, and environments, indicating that SGE does not require per-environment tuning to operate effectively.

A. Simulation Benchmark Evaluation

SGE was evaluated in the exploration benchmark environment [9]. Depth data was added to the robot’s camera, and forward-only driving was enforced since waypoint generation depends on camera-aligned perception. Perfect localization was assumed for all planners. The primary evaluation metric was the explored volume. Semantic segmentation used a UPerNet model [38] with a ResNet-50 backbone; the framework is agnostic to the segmentation model. The simulation model was trained on 1500 labeled images from the benchmark environments [9] (80/20 training-validation split), with eight classes mapped to the utility roles of (1) as a per-platform configuration: traversable (ground, trail, tunnel), small baseline utility (grass, vegetation), and obstacle (rock, wall, door). All simulations ran on a workstation with an i7-12700K CPU, 64 GB RAM, and an RTX 3090 GPU.

Figure 4 shows a representative exploration run of the SGE method in the campus environment. The robot trajectory is shown in green and the explored volume in blue. The campus layout contains elevation changes, multi-level passages, tunnels, and narrow entrances that are challenging for camera-based exploration. Our method successfully explored all accessible regions, achieving complete environment coverage.

Each method was evaluated five times in the campus environment. A run was considered complete if no further progress was observed for one minute, a one-hour time limit was reached, or the robot returned to its starting



Fig. 4: Representative exploration trajectory in the campus environment. Green: robot path. Blue: explored volume

position, indicating full coverage. As shown in Table I, SGE achieved an average explored volume of 44423m^3 , corresponding to complete exploration of the environment. This performance is comparable to TARE [24] and substantially exceeds NBVP [18] in this scenario. While SGE traversed a longer total distance than TARE, this behavior reflects its image-based waypoint generation, which produces shorter-range goals due to camera field-of-view and depth limits; the longer path also yields more fine-grained coverage and consistent exploration progression.

TABLE I: Exploration Benchmark in Simulation

Campus Environment		
Method	Volume (m^3)	Distance (m)
SGE	44423	3690
NBVP [18]	19980	377
TARE [24]	45875	2341

To evaluate the contribution of each SGE component, we conducted an ablation study following the same benchmark protocol (five runs per configuration, terminated at one hour or upon completion or failure). Results are summarized in Table II. Three observations emerge.

First, removing any individual component degrades mean explored volume relative to the full formulation, indicating that each contributes meaningfully. Removing the depth-based reward (c_d) reduced outward expansion, confirming its role as a surrogate for volumetric information gain. Eliminating obstacle repulsion (c_b) increased sampling near obstacles and reduced navigation reliability. Removing the spatial consistency term (c_a) increased sensitivity to segmentation noise, leading to less stable waypoint selection.

Second, ablating the utility terms sharply increases

TABLE II: Ablation Study in Campus Environment (mean $\pm$ std dev over 5 runs)

Variant	Volume (m^3)	Distance (m)	Runtime (s)
Full SGE	44423 $\pm$ 659	3690 $\pm$ 230	2633 $\pm$ 114
No Taboo Regions	42153 $\pm$ 3591	3526 $\pm$ 575	3204 $\pm$ 568
No Relocation	39216 $\pm$ 6897	3332 $\pm$ 793	2968 $\pm$ 880
No c_a (Spatial)	40370 $\pm$ 7274	3422 $\pm$ 1336	2768 $\pm$ 982
No c_b (Obstacle)	31621 $\pm$ 8408	2199 $\pm$ 915	1922 $\pm$ 834
No c_d (Depth)	27160 $\pm$ 13743	2394 $\pm$ 1706	2151 $\pm$ 1623

run-to-run variability. The full formulation achieved $44423 \pm 659 m^3$, whereas ablated variants exhibited standard deviations up to an order of magnitude larger (e.g., $\pm 13743 m^3$ without the depth term). The semantic utility terms therefore contribute not only to mean coverage but to the reliability and repeatability of exploration: consistent, complete coverage of the environment was achieved only with the full formulation.

Third, the Exploration Manager mechanisms provide fault tolerance under execution uncertainty. Disabling taboo regions or relocation each resulted in incomplete exploration in 2 of 5 trials. Without taboo regions, the system repeatedly attempted visually reachable but physically inaccessible waypoints (e.g., regions at different elevations or occluded by structural constraints), causing navigation stalls. Without relocation, the planner occasionally failed to transition between distant frontiers when the local planner could not complete long traversals. Together, these results indicate a division of roles: the semantic utility terms govern goal quality and consistency, while the manager mechanisms safeguard execution under navigation uncertainty.

Beyond volumetric performance, a key objective of SGE is controllable semantic biasing. To evaluate this capability, we conducted experiments in a modified forest environment [9] with labeled hiking trails [39]. Objects of interest are predefined semantic classes selected per mission, requiring no runtime annotation. Unlike purely geometric planners, SGE can bias sampling toward semantically relevant regions by adjusting the utility weights in (1): here, the trail class served as the object of interest and received slightly higher weights via c_a and c_c to favor sampling along trails.

Because the forest environment is an open-ended coverage problem, exploration time was limited to 1200s. This experiment demonstrates semantic adaptability rather than volumetric performance. As shown in Table III, SGE maintained similar traversal distance while allocating 39.46% of its trajectory along labeled trail regions. In contrast, TARE, which does not incorporate semantic awareness of trails, traversed only 6.77% of its path on trail. Although TARE explored slightly greater total volume in the allotted time, SGE demonstrated controllable semantic biasing without sacrificing exploration continuity.

TABLE III: Exploration Forest Trail Environment

Trail Forest Environment			
Method	Volume (m^3)	Distance (m)	Percentage on trail (%)
SGE	29629	1304	39.46
TARE	39905	1432	6.77

B. Real-World Exploration

SGE was validated on two ground robotic platforms across diverse environments, including an indoor campus building, an underground coal mine, and a limestone mine. Both platforms carried a 32-channel Velodyne Puck LiDAR; the Husky used a ZED2i stereo camera and Rhino an Intel RealSense D435i RGB-D camera. State estimation and mapping used LIO-SAM [40], and navigation the lattice-based FALCO planner [41]. The field segmentation model was trained on 2500 manually annotated images collected by the robots across the deployment environments (80/20 split), with ten classes: traversable (traversable terrain), small baseline utility (rough terrain, grass, vegetation), and obstacle (mud/water, rock, tree trunk, walls, doors, fixed structures). The SGE framework operated at the decision-making level, generating and managing exploration waypoints while relying on the underlying navigation stack for execution.

In the WVU Creative Arts Center (CAC), an indoor building with a non-standard circular layout, intersecting corridors, curved hallways, and pedestrian traffic, SGE successfully explored all accessible areas. The robot traversed 135m in 265s while executing 35 waypoints. Pedestrians were handled by the local navigation layer without modifying the exploration algorithm. Occasionally, temporarily open classroom doors resulted in waypoints being sampled inside rooms; when such goals proved unreachable, the taboo region mechanism excluded them, allowing exploration to continue. Figure 5(a) shows the Husky robot during exploration, and (b) the reconstructed map with explored volume (blue) and executed trajectory (green).

SGE was deployed in underground coal and limestone mine environments characterized by minimal illumination, dust, and irregular terrain. Figure 1(b) shows the onboard view from Rhino in a coal mine, with sampled image-space waypoints in blue. Sampled points avoid regions classified as mud and water, illustrating how semantic utility shapes waypoint selection and improves safety. Two trials were conducted in the East Fairfield limestone mine (Figure 5(c)), a large-scale environment with open spans up to 12m and 9×9 m pillar spacing. In the longest run, the robot operated for 2116s, traversed 765m, and explored $12835 m^3$. Figure 1(c) shows the resulting 3D reconstruction with the executed trajectory in green, highlighting the scale and structural complexity of the environment. A second trial achieved $10240 m^3$ over 533m in 1462s, with Figure 5(d) showing the corresponding top-down view. In both trials, exploration

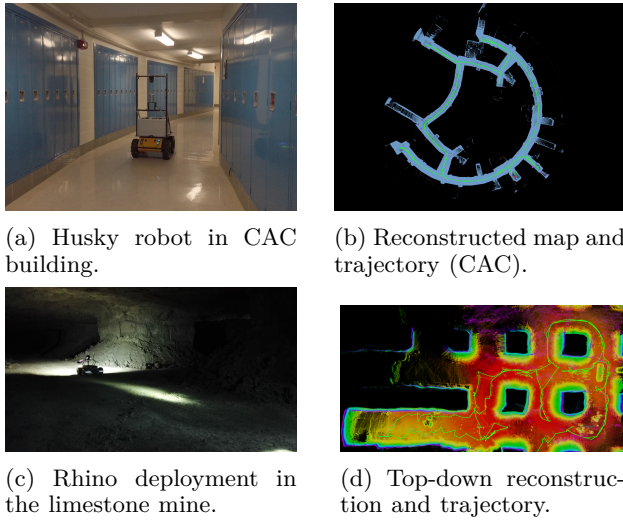


Fig. 5: Field deployments of SGE.

ended due to battery and hardware limits rather than algorithmic termination.

Despite segmentation uncertainty in low-visibility conditions, the Exploration Manager maintained stable behavior. Misclassified regions occasionally produced unreachable waypoints (e.g., near pillars), which the taboo mechanism excluded, preventing repeated attempts and allowing exploration to proceed. When the TSP optimizer selected distant frontiers, relocation maneuvers enabled structured backtracking through explored regions. These behaviors were consistent across deployments, indicating resilience to perception noise and navigation uncertainty.

V. Conclusion and Future Work

This work presented SGE, an exploration framework in which pixel-level semantics is the primary driver of waypoint generation. Across simulation benchmarks and field deployments, SGE achieved reliable large-scale exploration in real time, with volumetric coverage comparable to state-of-the-art geometric planners while additionally providing task-conditioned behavior that purely geometric methods do not, particularly in structurally complex environments.

Semantic-guided exploration represents an initial step toward fully learning-based exploration. Future work will investigate online terrain-aware cost adaptation, train visual attention-based networks for direct waypoint sampling from RGB-D inputs, and conduct broader evaluations in extreme terrains where hazardous and unpredictable conditions rigorously test robustness.

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